

RESEARCH INTERESTS

Machine Learning, Deep Learning, Multilingual and Multimodal NLP, Low-Resource Languages, AI Safety, Multimodal Learning, Clinical Decision Support Systems, and AI for Health Systems in LMICs.

EDUCATION

Brown University Ph.D. in Computational Biology, Advisor: Hamish Fraser M.S. in Computer Science, GPA: 3.9/4.0	Providence, USA 2023–Present 2021–2023
Nelson Mandela -AIST M.S. in Information and Communication Science and Engineering, GPA: 4.34/5.00	Arusha, Tanzania 2012–2014
Saint Augustine University of Tanzania B.S. in Computer Science, GPA: 4.50/5.00	Iringa, Tanzania 2006–2009

EXPERIENCE

Graduate Research Assistant Brown University	Providence, USA 2023–Present
- Conduct research on the evaluation, robustness, and safety of machine learning, deep learning, and large language model (LLM)–based clinical decision support systems. - Develop multilingual and low-resource evaluation benchmarks to assess reliability, hallucination risk, and policy compliance in healthcare settings.	
Data Scientist Center for Financial Inclusion (Cenfri)	Remote, South Africa March–November 2018
Led the development of a machine learning–based credit scoring system to improve financial access for underserved rural communities in Tanzania.	
Assistant Lecturer College of Business Education	Dodoma, Tanzania 2015–2021
- Taught undergraduate courses in programming (C/C++), business mathematics, and supervised capstone research projects. - Mentored students in applied machine learning and software development projects.	
Teaching Assistant Ruaha Catholic University	Iringa, Tanzania 2009–2010
Supported instruction in computer science courses and assisted with laboratory sessions and grading.	

PUBLICATIONS

- [1] T. Abdullahi, **M. Mgonzo**, M. Oduwale, P. Okewunmi, A. Owodunni, R. Singh, and C. Eickhoff, “Ubuntuguard: A culturally-grounded policy benchmark for equitable ai safety in african languages”, *AAAI Workshop on Language Models for Underserved Communities*, 2026.

- [2] **M. Mgonzo**, K. Oketch, N. A. Etori, W. Mang’eni, E. F. Nyaki, and M. S. Mollel, “Learning from scarcity: Building and benchmarking speech technology for Sukuma.”, in *Proceedings of the Second Workshop on Language Models for Low-Resource Languages (LoResLM 2026)*, Association for Computational Linguistics, 2026.
- [3] R. Esfandiarpour, G. Zerveas, R. Zhang, **M. Mgonzo**, C. Eickhoff, and S. Bach, “Beyond contrastive learning: Synthetic data enables list-wise training with multiple levels of relevance”, in *EMNLP*, 2025.

PROFESSIONAL SERVICE

- **Reviewing**
 - Machine Learning for Healthcare (ML4H) Conference 2023–2025
 - Black in AI (BAI) Workshop 2024
 - Deep Learning Indaba (DLI) Conference 2024, 2025
- **Volunteering**
 - SIGIR 2022
 - FAccT 2022
- **Organizing**
 - IndabaX, Nelson Mandela AIST, Arusha, Tanzania 2020

TEACHING

- **Graduate Teaching Assistant** at Brown University Fall 2024
Evaluation of Health Information Systems (BIOL 2075)
- **Assistant Lecturer** at College of Business Education 2010–2021
Advanced Business Mathematics, Programming in C, C++ (SMU 7304, ITT 5102 & ITT 6102)

SKILLS

- **Programming:** Python (advanced), C/C++, SQL.
- **ML & AI:** PyTorch, TensorFlow, HuggingFace, Scikit-learn.
- **Developer Tools:** Git, Linux, LaTeX.
- **Big Data Analytics Frameworks:** Hadoop, MapReduce.
- **Languages:** English Fluent; Swahili Native Speaker.

SELECTED PROJECTS

- **Cross-Modal Alignment for Unpaired Multimodal Data** 2025
Tools used: ViTs, BERT, Gromov-Wasserstein Optimal Transport
- **Systematicity in Recurrent Transformers** 2023
Tools used: Universal Transformer

SCHOLARSHIPS AND AWARDS

- Black in AI Travel Award (NeurIPS 2022) 2022
- Google CS Research Mentorship Program (CS-RMP) 2021
- ACM Travel Grant (FAccT 2019, FAccT 2020) 2019, 2020
- Google Conference Travel Award (DLI 2018) 2018